

PRIYANKA PATIL

Mechanical Design Engineer | Product Development Engineer
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PROFESSIONAL SUMMARY

Mechanical Design / Product Development Engineer with a Master's in Mechanical Engineering and ~3 years of combined design, manufacturing, and research experience across power-distribution, industrial-machinery, and semiconductor applications. Takes metal parts and mechanical/electromechanical assemblies from concept through design, prototyping, and release to manufacturing—applying GD&T (ASME Y14.5), tolerance stack-up analysis, and DFM/DFA to balance manufacturability, cost, and reliability. Skilled in 3D CAD modeling, 2D detailed drawings, FEA (ANSYS), FMEA/DFMEA root-cause analysis, BOM creation, and PLM-based design documentation. Proven record of accelerating design cycles, reducing prototype rework, and collaborating with machinists, manufacturing, procurement, and cross-functional teams to deliver production-ready designs.

CORE SKILLS

CAD & Drafting: Siemens NX, CATIA V5, SolidWorks, Creo, Solid Edge, AutoCAD; 3D modeling, 2D detailed/manufacturing drawings, assembly drawings, machine layouts

Standards & Tolerancing: GD&T (ASME Y14.5), tolerance stack-up analysis, drafting standards, dimensional inspection

Design & Manufacturing: DFM/DFA, sheet metal design, tooling & fixture design, material selection, prototype development & test iterations, manufacturing methods, CNC/precision machining knowledge, BOM; shop fundamentals—manual machining (lathe), welding, fabrication (academic)

Analysis & Simulation: FEA (ANSYS Mechanical – Static Structural), thermal-structural analysis, flow analysis, hand calculations, FMEA/DFMEA, parametric optimization, root cause analysis

PLM & Documentation: Siemens Teamcenter & Teamcenter Visualization (PLM/PDM), SAP (Material Master, BOMs), engineering change orders (ECO/ECN), design-data & BOM management, specification writing, technical documentation

Programming & Tools: Python, MATLAB, MS Excel (data analysis & reporting); Unity, Blender (AR/VR)

Collaboration & Leadership: Cross-functional teams, design reviews, technical presentations to teams/customers, supplier & manufacturer coordination, mentoring

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer (Intern), Low & Medium Voltage

May 2024 – Nov 2024

Siemens

Wendell, NC

- Designed electromechanical sub-assemblies for low- and medium-voltage power-distribution equipment in Siemens NX, applying GD&T (ASME Y14.5) and creating Bills of Materials (BOMs) and Material Master data to meet manufacturing and assembly requirements.
- Modified and refined CAD parts for medium-voltage switchgear (SIEBREAK)—redesigning busbars and switchgear door components with added alignment features—and released production-ready designs to manufacturing, improving assembly fit, reducing installation errors, and increasing production consistency.
- Assembled and integrated low-voltage electrical and mechanical components on switchboard panels to validate component fit, alignment, and assembly requirements.
- Owned engineering change orders (ECO/ECN), managed design data, component libraries, and BOM revisions in Siemens Teamcenter (PLM), and authored detailed work instructions and process documentation to ensure standardization and efficient design reuse.

Mechanical Design Engineer

Jan 2023 – Jul 2023

Laxmi Techno Works

Pune, India

- Owned the end-to-end design cycle for custom industrial machines—gathering requirements directly from customers, creating new and client-specific designs in Creo (3D part/assembly models and GD&T-based 2D manufacturing drawings), and following them from concept through release to built, delivered machines.
- Collaborated directly with vendors on design-for-manufacturability (DFM) feedback and managed engineering change orders (ECO/ECN) through release.
- Accelerated design iteration cycles by ~15% by standardizing design templates, tightening requirement-to-drawing handoffs, and running rapid design-review loops.
- Reduced prototype rework by ~12% and improved dimensional compliance by ~10% through tolerance stack-up and fit/clearance checks, simulation-driven validation, and structured root-cause analysis (DFMEA) across build-test iterations.
- Shortened fabrication lead time by ~2 days by aligning tooling/fixture concepts and component selection with manufacturing and procurement teams.

Mechanical Engineer (Intern)

EAMS Engineering & Management India Pvt. Ltd.

Feb 2021 – Jul 2021

Pune, India

- Optimized manufacturing line balancing and capacity planning and built a centralized documentation framework that improved operational transparency across cross-functional teams.

Mechanical Engineer (Intern)

Solitaire Engineers

May 2019 – Jun 2019

Pune, India

- Created GD&T-based production drawings in AutoCAD with machining process documentation and performed dimensional inspections on 20+ components.

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Cincinnati

Feb 2025 – Present

Cincinnati, OH

- Developed a 1:1 scale 3D CAD model of Atomic Layer Deposition (semiconductor) equipment in Siemens NX, building mechanical assemblies and assembly animations in NX and Blender for interactive AR/VR training systems.
- Built AR-based mechanical assembly guidance and component-recognition tooling (custom-trained YOLO v8, ~95% test accuracy) supporting assembly training and maintenance.

ACADEMIC PROJECTS

Thermoelectric Generator Integration in Jet Engine

Sep 2023 – Nov 2023

- Performed thermal-structural design analysis and parametric optimization of a thermoelectric generator (TEG), tuning thickness and operating parameters under thermal-conductivity, surface-area, and heat-transfer constraints to achieve ~6–7% conversion efficiency.

Blood Plasma Separation Using Microfluidics

Sep 2023 – Nov 2023

- Validated geometry-driven separation performance through flow analysis in ANSYS using hex-dominant meshing, achieving numerical convergence (1e-11) and confirming feasibility of the micro flow-path design.

Comparative Study of Biomaterials — Femur Bone Case Study

Jan 2024 – Apr 2024

- Analyzed the mechanical behavior of femur biomaterials, comparing Co–Cr alloys and PMMA on load-carrying capacity, stress distribution, and manufacturability to guide material selection.

Non-Traditional Machining Processes in Biomedical Applications

Jan 2024 – Apr 2024

- Evaluated manufacturing trade-offs across four non-traditional machining processes (water-jet, ultrasonic, laser-beam machining, and electrocautery), comparing thermal effects, dimensional accuracy, and surface integrity to support precision-driven design decisions.

CERTIFICATIONS

Siemens NX Mastery: Advanced Design & Applications • AI for Design & Optimization • Generative AI: Introduction & Applications • Python for Data Science

EDUCATION

Master of Engineering in Mechanical Engineering

University of Cincinnati

Dec 2024

Cincinnati, OH, USA

Bachelor of Engineering in Mechanical Engineering

Maharashtra Institute of Technology

Apr 2020

Pune, India